

# Wired-to-Wireless RSSI Emulation & Calibration

Frequency Range: 220 MHz, 429 MHz, 920 MHz

Scope: 0-10 Meters (Near-to-Mid Field)

## 1. Overview

The goal of this method is to use a Wired Setup (Coaxial Cables + Variable Attenuator) to predict the Wireless RSSI of an ideal outdoor environment where the First Fresnel Zone (FFZ) is completely clear.

## 2. Theoretical Foundation

To make the wired signal equivalent to the wireless signal at distance  $d$ , the Attenuator (VATT) must account for Free Space Path Loss (FSPL) while compensating for the absence of physical antennas.

The Master Equation:

$$VATT_{\text{setting}} = FSPL(d, f) - (G_{\text{TX}} + G_{\text{RX}}) - L_{\text{cable}}$$

- $FSPL(d, f)$ : Free Space Path Loss at distance  $d$  and frequency  $f$ .
- $G_{\text{TX}} + G_{\text{RX}}$ : Combined Gain of your antennas (dBi). **Crucial: You must subtract this because the wired setup bypasses the antenna's physical amplification.**
- $L_{\text{cable}}$ : Insertion loss of the coaxial cables and connectors in your test bench.

## 3. Reference Table: Optimal Benchmarking Points

For outdoor validation, use these "Golden Points" where the wireless signal is most stable and closest to the theoretical model.

| Frequency<br>( $f$ ) | Wavelength<br>( $\lambda$ ) | Recom. Dist.<br>( $d$ ) | Mini. Height<br>( $h$ ) | $FSPL$ at 1 M<br>( $FSPL_0$ ) |
|----------------------|-----------------------------|-------------------------|-------------------------|-------------------------------|
| 220 MHz              | 1.36 m                      | 5.0 m                   | > 2.5 m                 | 19.3 dB                       |
| 429 MHz              | 0.70 m                      | 3.0 m                   | > 1.8 m                 | 25.1 dB                       |
| 920 MHz              | 0.33 m                      | 2.0 m                   | > 1.5 m                 | 31.7 dB                       |

## 4. Operational Steps

### Step A: Indoor Wired Setup (Ground Truth)

1. Connect the Transmitter (TX) to the Receiver (RX) using:  
[TX] -> [Cable 1] -> [Step Attenuator (VATT)] -> [Cable 2] -> [RX]
2. Calculate the target  $FSPL(d)$  for your desired distance.  
$$FSPL(d) = FSPL_0 + 20 \log_{10}(d)$$
3. Adjust the VATT using the Master Equation in Section 2.
4. Record the resulting RSSI. This is your **Predicted Ideal RSSI** for distance  $d$ .

### Step B: Outdoor Wireless Validation

1. Mount antennas on non-metallic stands (PVC or Wood).
2. Set the height ( $h$ ) according to the Reference Table to clear the First Fresnel Zone.
3. Ensure the polarization (Vertical/Horizontal) is identical for both ends.
4. Measure RSSI at the same distance  $d$ .
5. Comparison: If  $RSSI_{\text{Wireless}} \approx RSSI_{\text{Wired}}$ , your prediction model is calibrated.

## 5. Critical Constraints & Troubleshooting

- **The 220MHz Height Rule:** At 220MHz, the wavelength is long (~1.36m). If your antenna is lower than 2m, ground reflections will cause "multi-path fading," making your wireless RSSI significantly lower than the wired prediction.
- **Near-Field Warning:** Do not attempt wireless benchmarking at  $d < 1$  meter for these frequencies. The antennas are in the "induction near-field," where gain values are inconsistent.
- **Cable Loss Calibration:** Always measure your cable loss ( $L_{\text{cable}}$ ) at the specific frequency before starting, as it increases with frequency (e.g., loss at 920MHz is higher than at 220MHz).